



HK IMO Training 2019-2020
Problem Set 7 Suggested Solutions



Problem 25. There are 315 marbles divided into three piles of 81, 115 and 119. In each move Pythagoras can either merge several piles into a single pile or divide a pile with an even number of marbles into 2 equal piles. Can Pythagoras divide the marbles into 315 piles, each with a single marble?

Solution. The answer is no. First observe that when all the piles are multiples of an odd prime p , then after the next move, all the piles are still multiples of p . Since at the start, all piles are odd, Pythagoras can only merge two of the piles (if all piles are merged, no more moves can be carried out), ending with one of the following pairs: (81, 234), (115, 200), (119, 196), which are multiples of 3, 5 and 7 respectively. Therefore it is not possible to obtain a pile with a single marble.

Problem 26. Each point A in the plane is assigned a real number $f(A)$. It is known that

$$f(M) = f(A) + f(B) + f(C)$$

whenever M is the centroid of $\triangle ABC$. Prove that $f(A) = 0$ for every point A .

Solution. Let M be the centroid of $\triangle ABC$, and let D, E , and F be the centroids of $\triangle BCM$, $\triangle CAM$, and $\triangle ABM$, respectively. Then M is also the centroid of $\triangle DEF$. Thus,

$$\begin{aligned} f(M) &= f(D) + f(E) + f(F) \\ &= (f(B) + f(C) + f(M)) + (f(C) + f(A) + f(M)) + (f(A) + f(B) + f(M)) \\ &= 2(f(A) + f(B) + f(C)) + 3f(M) \\ &= 2f(M) + 3f(M) \\ &= 5f(M). \end{aligned}$$

This yields $f(M) = 0$. As every point is the centroid of some triangle, this shows $f(A) = 0$ for any A .

Problem 27. Find all positive integers d with the following property: there exists integer $k \geq 3$ such that the numbers $d, 2d, 3d, \dots, kd$ can be ordered in a sequence such that the sum of every two consecutive terms is a perfect square.

Solution. The answer is every positive integer d which is a perfect square.

Note that for any ordering of $d, 2d, 3d, \dots, kd$, the sum of any two consecutive terms must be a multiple of d . Therefore, we may remove all square factors of d and assume d is a squarefree positive integer.

For $d = 1$, we can take $k = 15$ and the sequence

$$8, 1, 15, 10, 6, 3, 13, 12, 4, 5, 11, 14, 2, 7, 9.$$

The sums of two consecutive terms are 9, 16, 25, 16, 9, 16, 25, 16, 9, 16, 25, 16, 9, 16, which are all perfect squares. Therefore, d can be any perfect square.

Now let $d > 1$. Suppose there exists such a permutation $a_1d, a_2d, \dots, a_kd$. As $a_id + a_{i+1}d = (a_i + a_{i+1})d$ is a square and d is squarefree, $a_i + a_{i+1}$ must be a multiple of d for each i . Thus,

$$a_1 \equiv -a_2 \equiv a_3 \equiv -a_4 \equiv \dots \pmod{d}.$$

In other words, $a_m \equiv (-1)^{m+1}a_1 \pmod{d}$. This means there are exactly two remainders modulo d among $1, 2, \dots, k$. As $k \geq 3$, the only possibility is $d = 2$. However, this yields $a_m \equiv a_1 \pmod{d}$ for any m , which is also a contradiction. This completes the proof.

Problem 28. Let ABC be an acute-angled triangle. Denote, respectively, by

- k_a, k_b , and k_c the lengths of the symmedians from A, B and C to BC, CA and AB ;
- r_a, r_b , and r_c the exradii of the excircles opposite A, B and C ;
- R and r the circumradius and inradius of $\triangle ABC$.

Prove that

$$k_a(r + r_a) + k_b(r + r_b) + k_c(r + r_c) \leq 4(R + r)^2.$$

Solution. Let $BC = a, CA = b$ and $AB = c$, let m_a, m_b, m_c be the lengths of the medians from A, B and C to BC, CA and AB , respectively. We will use the following facts:

- $k_a = \frac{2bcm_a}{b^2 + c^2}, k_b = \frac{2cam_b}{c^2 + a^2}, k_c = \frac{2abm_c}{a^2 + b^2}$ (by Stewart's Theorem and Menelaus's Theorem),

- $k_a \leq m_a \leq R(1 + \cos A)$, $k_b \leq m_b \leq R(1 + \cos B)$, $k_c \leq m_c \leq R(1 + \cos C)$ (by the AM-GM Inequality and the Triangle Inequality),
- $r_a(s - a) = r_b(s - b) = r_c(s - c) = rs = \Delta$, where s and Δ are respectively, the semi-perimeter and area of $\triangle ABC$.

Therefore, by Gerretsen's Inequality,

$$\begin{aligned}
k_a(r + r_a) + k_b(r + r_b) + k_c(r + r_c) &\leq R(1 + \cos A) \left(\frac{\Delta}{s} + \frac{\Delta}{s - a} \right) + R(1 + \cos B) \left(\frac{\Delta}{s} + \frac{\Delta}{s - b} \right) + R(1 + \cos C) \left(\frac{\Delta}{s} + \frac{\Delta}{s - c} \right) \\
&= 2Rrs \left(\frac{s(s - a)}{bc} \cdot \frac{b + c}{s(s - a)} + \frac{s(s - b)}{ca} \cdot \frac{c + a}{s(s - b)} + \frac{s(s - c)}{ab} \cdot \frac{a + b}{s(s - c)} \right) \\
&= 2Rrs \left(\frac{a(b + c)}{abc} + \frac{b(c + a)}{abc} + \frac{c(a + b)}{abc} \right) \\
&= \frac{4Rrs}{4Rrs} (ab + bc + ca) \\
&= s^2 + 4Rr + r^2 \\
&\leq (4R^2 + 4Rr + 3r^2) + 4Rr + r^2 \\
&= 4(R + r)^2.
\end{aligned}$$